

100 Days of Linux - Day 067

Title: Repeating Tasks with for and while Loops

Today's Goal

Learn how to repeat commands automatically using **for** and **while** loops in Bash.

Why This Matters

Loops are used to process files, automate backups, run repetitive tasks, and manage large numbers of resources in Linux and data engineering workflows.

Concepts of the Day

For Loop:

```
for ITEM in one two three
do
echo $ITEM
done
```

While Loop:

```
COUNT=1
while [ $COUNT -le 5 ]
do
echo $COUNT
COUNT=$((COUNT+1))
done
```

Arithmetic:

```
$((COUNT+1))
```

Beginner Lesson

A loop repeats the same block of commands multiple times. Use a **for** loop when you already know the items to process. Use a **while** loop when you want to continue until a condition becomes false.

Practice Questions

1. Create a script named `for_loop.sh` that prints the numbers 1 to 5 using a for loop.
2. Modify the script to print the names Ubuntu, Fedora, Debian, and Kali using a for loop.
3. Create a script named `while_loop.sh` that prints the numbers 1 to 5 using a while loop.
4. Create a script that loops through all `.txt` files in the current directory and prints each filename.
5. Run your scripts and display your current working directory.

Hints

Remember to close loops with done. Use `$(( ))` for arithmetic in Bash.

Mini Challenge

Create five sample text files. Write a script that displays the name of each file and counts from 1 to 5 before printing 'Task

Complete'.

Common Mistakes

Forgetting done, not updating the counter in a while loop, or using incorrect arithmetic syntax.

Did You Know?

Many system administration scripts use loops to process hundreds or even thousands of files automatically.

Pro Tip

Test loops with a small number of items before using them on large directories or production data.

Answer Key (Instructor Only)

Q1: for i in 1 2 3 4 5; do echo \$i; done

Q2: for os in Ubuntu Fedora Debian Kali; do echo \$os; done

**Q3: COUNT=1; while [\$COUNT -le 5]; do echo \$COUNT;
COUNT=\$((COUNT+1)); done**

Q4: for f in *.txt; do echo \$f; done

Q5: bash for_loop.sh && bash while_loop.sh && pwd

Homework

Create two additional scripts: one that loops through all .sh files and another that prints the days of the week using a for loop.

Next Day Preview

Tomorrow you'll learn Bash functions to organize reusable blocks of code and make scripts cleaner and easier to maintain.